



# DIFFERENCES BETWEEN

## SOAP and REST

SOAP



VS



REST

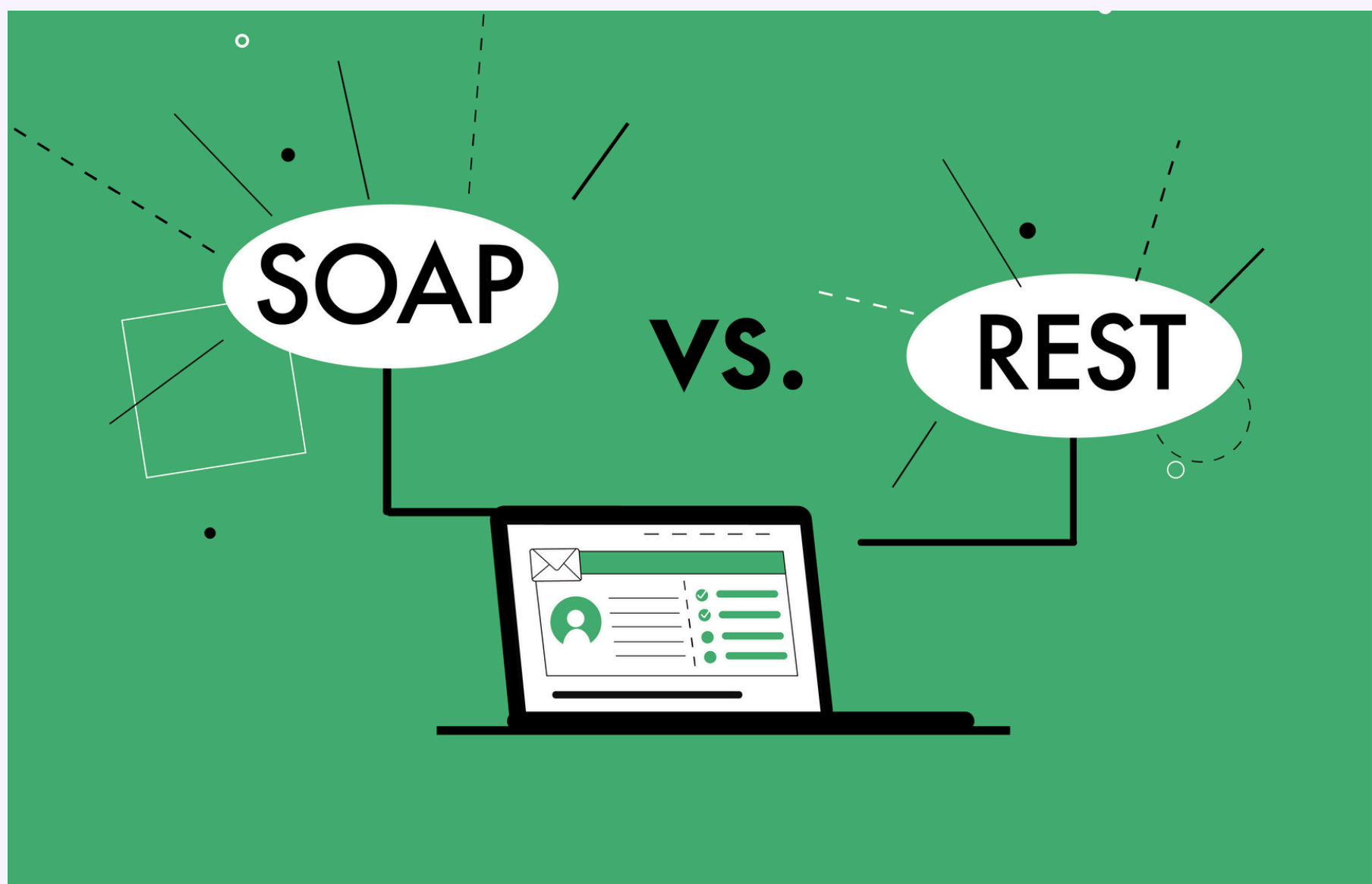




# FULL FORMS:

---

- **SOAP:** Simple Object Access Protocol.
- **REST:** Representational State Transfer.

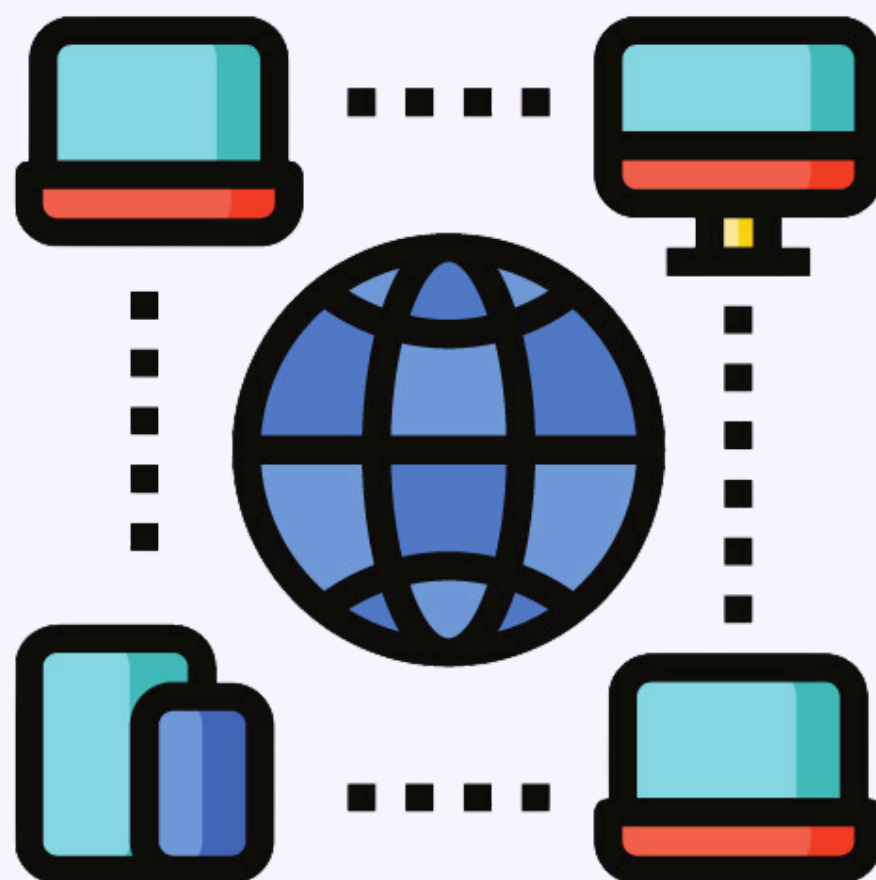




# PROTOCOL:

---

- **SOAP:** A protocol with strict standards and rules.
- **REST:** An architectural style that uses existing protocols, typically HTTP.





# MESSAGE FORMAT:

---

- **SOAP:** Uses XML for message format.
- **REST:** Can use multiple formats like JSON, XML, HTML, and plain text.

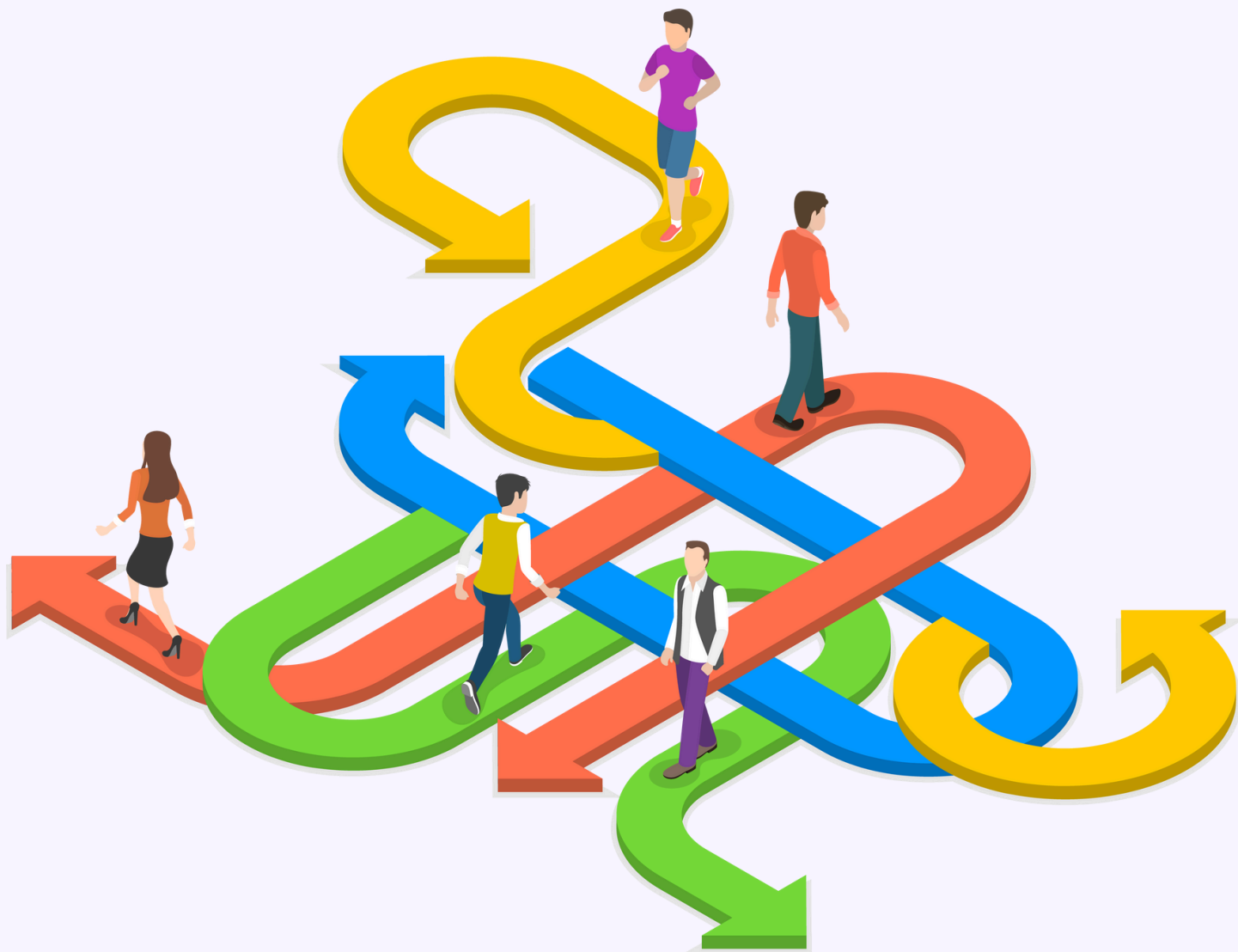




# COMPLEXITY:

---

- **SOAP:** More complex due to strict standards and extensive features.
- **REST:** Simpler and more flexible.

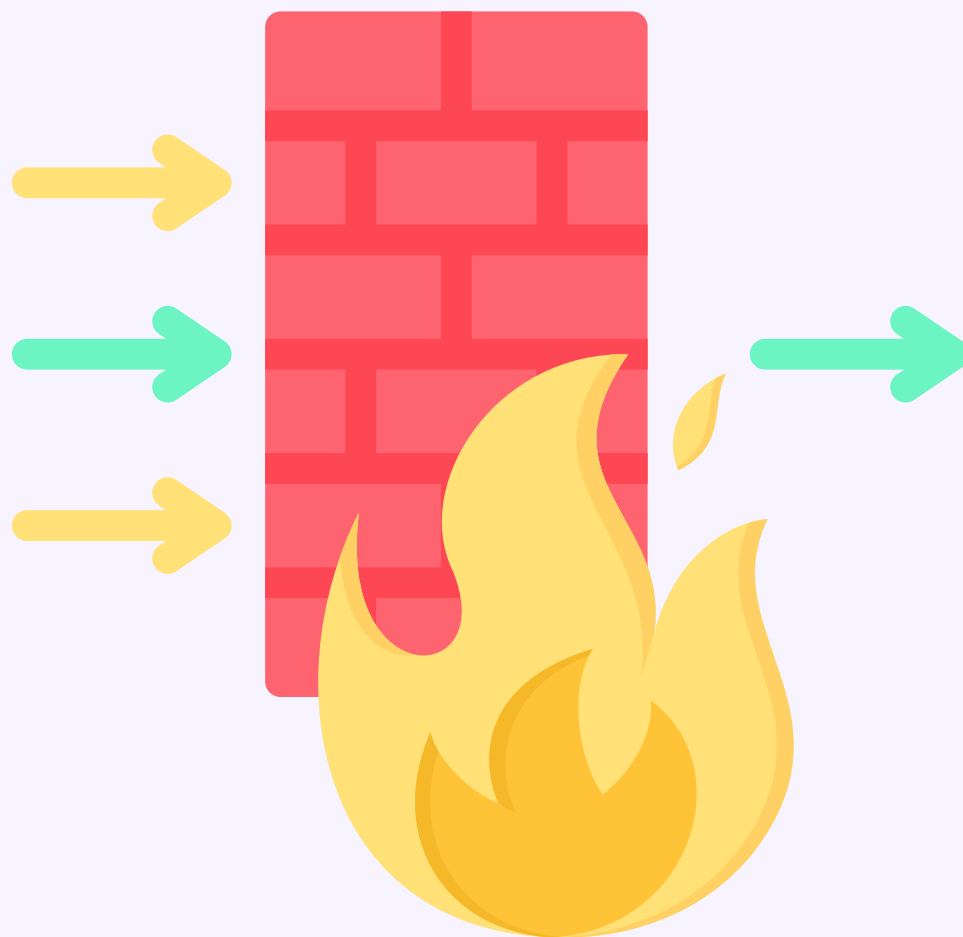




# STATEFULNESS:

---

- **SOAP:** Can be stateful (maintains state between requests).
- **REST:** Stateless (each request from a client contains all the information needed).





# PERFORMANCE:

---

- **SOAP:** Generally slower due to its complexity and extensive use of XML.
- **REST:** Faster and more efficient, especially with JSON.





# ERROR HANDLING:

---

- **SOAP:** Provides built-in error handling mechanisms through SOAP fault.
- **REST:** Relies on HTTP status codes for error handling.





# SECURITY:

---

- **SOAP:** Built-in security features like WS-Security.
- **REST:** Security is implemented through underlying protocols like HTTPS.





# TRANSPORT PROTOCOLS:

---

- **SOAP:** Can work over any protocol such as HTTP, SMTP, TCP, etc.
- **REST:** Typically works over HTTP/HTTPS.

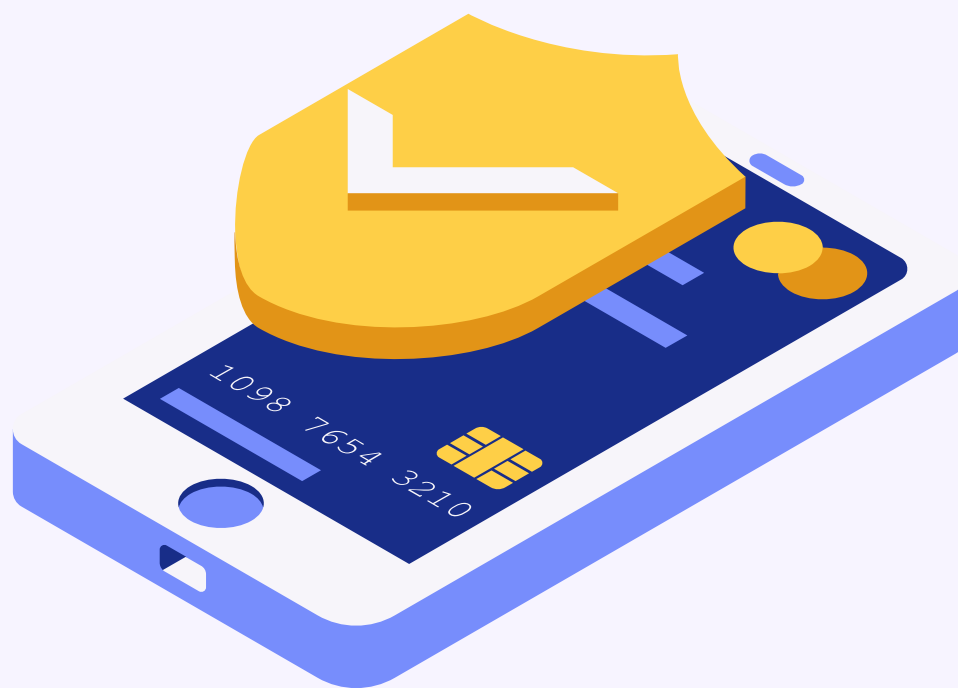




# USE CASES:

---

- **SOAP:** Preferred for enterprise-level services requiring high security and transactional reliability.
- **REST:** Widely used for web services requiring scalability and performance.





# STANDARDS AND EXTENSIBILITY:

---

- **SOAP:** Highly standardized with built-in extensibility through protocols like WS-Addressing, WS-ReliableMessaging.
- **REST:** Less standardized but more adaptable, using URIs, HTTP methods, and headers to achieve various functionalities.





**Follow us to get more  
Information  
and tips like this.**



@MYJOBGROW